

Brutal Practice Paper B47 - Nutrition in Plants, Comparing Quantities, Winds & Cyclones, Lines & Angles

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A leaf makes 6 g of glucose for each hour of good sunlight by photosynthesis. On a cloudy day it gets only 4 hours of good sunlight instead of its usual 9. How much less glucose does it make that day? [\[Nutrition in](#)

[Plants\]](#)

- (A) 24 g
- (B) 42 g
- (C) 48 g
- (D) 30 g
- (E) 78 g
- (F) 54 g
- (G) 5 g

2. A recipe mixes sugar and flour in the ratio 2 : 5. To make a batch using 350 g of flour, how much sugar is needed? [\[Comparing Quantities\]](#)

- (A) 100 g
- (B) 175 g
- (C) 700 g
- (D) 140 g
- (E) 210 g
- (F) 250 g
- (G) 87.5 g

3. Warm air rises because heating makes it expand and grow lighter. A trapped pocket of air, 5 litres at room temperature, is warmed until its volume increases by 20 percent. What is its new volume? [\[Winds, Storms](#)

[& Cyclones\]](#)

- (A) 6 litres
- (B) 1 litre
- (C) 5.2 litres
- (D) 10 litres
- (E) 7 litres
- (F) 25 litres
- (G) 4 litres

4. Two angles are complementary, and one of them is twice the other. What is the measure of the larger angle? [\[Lines & Angles\]](#)

- (A) 80 degrees
- (B) 30 degrees
- (C) 90 degrees
- (D) 60 degrees
- (E) 45 degrees
- (F) 75 degrees

5. In photosynthesis, 6 molecules of carbon dioxide make 1 molecule of glucose. If a plant uses up 30 molecules of carbon dioxide, how many glucose molecules can it build? [\[Nutrition in Plants\]](#)

- (A) 30
- (B) 180
- (C) 36
- (D) 6
- (E) 25
- (F) 5
- (G) 12 molecules

6. A shopkeeper buys a bag for Rs 480 and sells it at a profit of 25 percent. What is the selling price?

[\[Comparing Quantities\]](#)

- (A) Rs 120
- (B) Rs 600
- (C) Rs 560
- (D) Rs 480
- (E) Rs 605
- (F) Rs 720
- (G) Rs 505

7. A cyclone's winds carry it 240 km in 4 hours. What is the speed of the cyclone in kilometres per hour?

[\[Winds, Storms & Cyclones\]](#)

- (A) 80 km/h
- (B) 60 km/h
- (C) 24 km/h
- (D) 120 km/h
- (E) 100 km/h
- (F) 960 km/h
- (G) 40 km/h

8. An angle is 40 degrees less than its own supplement. What is the measure of the angle? [\[Lines & Angles\]](#)

- (A) 100 degrees
- (B) 55 degrees
- (C) 110 degrees
- (D) 140 degrees
- (E) 70 degrees
- (F) 90 degrees
- (G) 35 degrees

9. Tiny pores called stomata cover a leaf, about 250 in every square millimetre. How many stomata are on a rectangular patch measuring 3 mm by 4 mm? [\[Nutrition in Plants\]](#)

- (A) 250
- (B) 3500
- (C) 1000
- (D) 2500
- (E) 3000
- (F) 750
- (G) 1200

10. A jacket is marked at Rs 2000 and sold at a 15 percent discount. What is the sale price? [\[Comparing Quantities\]](#)

- (A) Rs 1700
- (B) Rs 300
- (C) Rs 1985
- (D) Rs 1715
- (E) Rs 2300
- (F) Rs 1850

11. Wind blows from high pressure to low pressure. Station A reads 1018 units of pressure and station B reads 994 units. What is the pressure difference that drives the wind between them? [\[Winds, Storms & Cyclones\]](#)

- (A) 1006 units
- (B) 24 units
- (C) 2012 units
- (D) 20 units
- (E) 34 units
- (F) 14 units

12. Two angles form a linear pair, and one is 5 times the other. What is the measure of the smaller angle? [\[Lines & Angles\]](#)

- (A) 20 degrees
- (B) 90 degrees
- (C) 36 degrees
- (D) 150 degrees
- (E) 45 degrees
- (F) 30 degrees

13. A variegated leaf is part green and part white, and only the green parts make food. If 60 percent of the leaf's 40 square centimetres is green, what area can photosynthesise? [\[Nutrition in Plants\]](#)

- (A) 16 square centimetres
- (B) 30 square centimetres
- (C) 4 square centimetres
- (D) 24 square centimetres
- (E) 40 square centimetres
- (F) 10 square centimetres
- (G) 20 square centimetres

14. Find the simple interest on Rs 5000 borrowed at 8 percent per year for 3 years. [\[Comparing Quantities\]](#)

- (A) Rs 1200
- (B) Rs 400
- (C) Rs 6200
- (D) Rs 800
- (E) Rs 4000
- (F) Rs 1500
- (G) Rs 1320

- 15.** A cyclone is moving towards the coast at 15 km/h and is now 90 km out at sea. In how many hours will it reach the coast? [\[Winds, Storms & Cyclones\]](#)
- (A) 7 hours
 - (B) 5 hours
 - (C) 105 hours
 - (D) 6 hours
 - (E) 90 hours
 - (F) 75 hours
 - (G) 1350 hours
- 16.** Two straight lines cross each other. One of the four angles formed is 110 degrees. What is the measure of the angle next to it (its adjacent angle)? [\[Lines & Angles\]](#)
- (A) 55 degrees
 - (B) 20 degrees
 - (C) 250 degrees
 - (D) 90 degrees
 - (E) 100 degrees
 - (F) 70 degrees
 - (G) 110 degrees
- 17.** Amarbel (*Cuscuta*) is a parasite plant with no chlorophyll, so it steals food from its host. If it covers one quarter of a hedge 60 m long, how many metres of the hedge are free of the parasite? [\[Nutrition in Plants\]](#)
- (A) 35 m
 - (B) 50 m
 - (C) 20 m
 - (D) 40 m
 - (E) 45 m
 - (F) 15 m
 - (G) 30 m
- 18.** In a class, 60 percent of the students are girls. Of those girls, 25 percent wear glasses. What percent of the whole class are girls who wear glasses? [\[Comparing Quantities\]](#)
- (A) 12 percent
 - (B) 85 percent
 - (C) 35 percent
 - (D) 15 percent
 - (E) 40 percent
 - (F) 25 percent
 - (G) 20 percent
- 19.** During a storm the wind speed climbs from 40 km/h to 100 km/h. What is the percentage increase in wind speed? [\[Winds, Storms & Cyclones\]](#)
- (A) 100 percent
 - (B) 60 percent
 - (C) 40 percent
 - (D) 60
 - (E) 150 percent
 - (F) 300 percent

- 20.** A right angle is divided into two complementary angles whose measures are in the ratio 2 : 3. What is the larger of these two angles? [\[Lines & Angles\]](#)
- (A) 72 degrees
 - (B) 54 degrees
 - (C) 36 degrees
 - (D) 60 degrees
 - (E) 45 degrees
 - (F) 108 degrees
 - (G) 30 degrees
- 21.** A pitcher plant traps insects to get the nitrogen its boggy soil lacks. If it catches 8 insects each day, how many will it trap over the 3 weeks of an experiment? [\[Nutrition in Plants\]](#)
- (A) 24
 - (B) 176
 - (C) 21
 - (D) 160
 - (E) 56
 - (F) 168
 - (G) 32
- 22.** Rs 1800 is shared between two friends in the ratio 5 : 4. How much does the friend with the larger share receive? [\[Comparing Quantities\]](#)
- (A) Rs 1200
 - (B) Rs 800
 - (C) Rs 1000
 - (D) Rs 900
 - (E) Rs 1080
 - (F) Rs 500
- 23.** Monsoon winds bring 80 percent of a region's yearly rain, which totals 1500 mm. How much rain falls outside the monsoon season? [\[Winds, Storms & Cyclones\]](#)
- (A) 375 mm
 - (B) 750 mm
 - (C) 1500 mm
 - (D) 80 mm
 - (E) 300 mm
 - (F) 200 mm
 - (G) 1200 mm
- 24.** A transversal cuts two parallel lines. One of a pair of co-interior (same-side interior) angles is 65 degrees. What is the other co-interior angle? [\[Lines & Angles\]](#)
- (A) 180 degrees
 - (B) 95 degrees
 - (C) 100 degrees
 - (D) 130 degrees
 - (E) 65 degrees
 - (F) 115 degrees
 - (G) 25 degrees

- 25.** Bread mould is a saprotroph that doubles the area it covers every day. If it covers 2 square centimetres on day 1, what area does it cover on day 4? [\[Nutrition in Plants\]](#)
- (A) 16 square centimetres
 - (B) 8 square centimetres
 - (C) 14 square centimetres
 - (D) 32 square centimetres
 - (E) 24 square centimetres
 - (F) 10 square centimetres
 - (G) 6 square centimetres
- 26.** If 8 identical pens cost Rs 96, what will 15 such pens cost at the same rate? [\[Comparing Quantities\]](#)
- (A) Rs 180
 - (B) Rs 160
 - (C) Rs 12
 - (D) Rs 720
 - (E) Rs 144
 - (F) Rs 200
- 27.** The cups of an anemometer spin 5 times in 2 seconds in a steady wind. How many turns will they make in one full minute? [\[Winds, Storms & Cyclones\]](#)
- (A) 75
 - (B) 300
 - (C) 60
 - (D) 2.5
 - (E) 120
 - (F) 150
 - (G) 10
- 28.** Three angles lie next to each other on a straight line and measure x , $2x$ and $3x$. What is the value of x ? [\[Lines & Angles\]](#)
- (A) 90 degrees
 - (B) 36 degrees
 - (C) 60 degrees
 - (D) 30 degrees
 - (E) 45 degrees
 - (F) 18 degrees
- 29.** Rhizobium bacteria living in the root nodules of a pulse crop fix nitrogen from the air. A field needs 90 kg of nitrogen per hectare, and Rhizobium supplies two thirds of it. How much must the farmer add as fertiliser? [\[Nutrition in Plants\]](#)
- (A) 90 kg
 - (B) 15 kg
 - (C) 70 kg
 - (D) 60 kg
 - (E) 45 kg
 - (F) 30 kg

30. Over one month a sapling's height went up from 40 cm to 50 cm. By what percentage did its height increase? [\[Comparing Quantities\]](#)

- (A) 25 percent
- (B) 30 percent
- (C) 80 percent
- (D) 15 percent
- (E) 125 percent
- (F) 10 percent

31. A steady wind blows at 72 km/h. How many metres does it travel in just one second? [\[Winds, Storms & Cyclones\]](#)

- (A) 72 m
- (B) 2 m
- (C) 36 m
- (D) 200 m
- (E) 12 m
- (F) 20 m
- (G) 1.2 m

32. An angle and its complement are in the ratio 4 : 5. What is the measure of the smaller angle? [\[Lines & Angles\]](#)

- (A) 45 degrees
- (B) 40 degrees
- (C) 36 degrees
- (D) 50 degrees
- (E) 80 degrees
- (F) 30 degrees
- (G) 72 degrees

33. A farmer rotates crops: a cereal that removes 50 kg of nitrogen from the soil, then a pulse that puts 30 kg back. After one full cycle, what is the net change in the soil's nitrogen? [\[Nutrition in Plants\]](#)

- (A) a gain of 30 kg
- (B) a gain of 80 kg
- (C) a loss of 30 kg
- (D) a loss of 80 kg
- (E) a loss of 20 kg
- (F) a loss of 50 kg
- (G) a gain of 20 kg

34. A trader buys oranges for Rs 600 and sells them all for Rs 510. Find the loss percent. [\[Comparing Quantities\]](#)

- (A) 12 percent
- (B) 90 percent
- (C) 15 percent
- (D) 17.6 percent
- (E) 18 percent
- (F) 10 percent
- (G) 9 percent

- 35.** Land heats faster than the sea, so cooler sea air rushes in as a sea breeze. If the breeze covers 9 km in 30 minutes, what is its speed in km/h? [\[Winds, Storms & Cyclones\]](#)
- (A) 36 km/h
 - (B) 18 km/h
 - (C) 9 km/h
 - (D) 13.5 km/h
 - (E) 30 km/h
 - (F) 270 km/h
 - (G) 4.5 km/h
- 36.** A transversal cuts two parallel lines so that a pair of alternate interior angles measure $(2x + 10)$ degrees and $(x + 40)$ degrees. What is the value of x ? [\[Lines & Angles\]](#)
- (A) 20
 - (B) 50
 - (C) 40
 - (D) 30
 - (E) 60
 - (F) 10
- 37.** A water plant releases 3 oxygen bubbles a minute under one lamp. Under two lamps the rate triples. How many bubbles does it release in 5 minutes under two lamps? [\[Nutrition in Plants\]](#)
- (A) 9
 - (B) 45
 - (C) 60
 - (D) 30
 - (E) 75 bubbles
 - (F) 15
 - (G) 18
- 38.** In a school the ratio of teachers to students is 1 : 24. What percent of all the people in the school are teachers? [\[Comparing Quantities\]](#)
- (A) 8 percent
 - (B) 2 percent
 - (C) 6 percent
 - (D) 4 percent
 - (E) 20 percent
 - (F) 24 percent
 - (G) 5 percent
- 39.** The calm centre, or eye, of a cyclone is 30 km across, while the whole storm is 450 km across. The eye's width is what fraction of the storm's width, in lowest terms? [\[Winds, Storms & Cyclones\]](#)
- (A) $\frac{1}{9}$
 - (B) $\frac{2}{15}$
 - (C) $\frac{1}{10}$
 - (D) $\frac{1}{15}$
 - (E) $\frac{1}{12}$
 - (F) $\frac{1}{30}$
 - (G) $\frac{1}{5}$

40. Two supplementary angles differ from each other by 30 degrees. What is the measure of the larger angle? [\[Lines & Angles\]](#)

- (A) 75 degrees
- (B) 105 degrees
- (C) 60 degrees
- (D) 90 degrees
- (E) 135 degrees
- (F) 150 degrees

41. Iodine turns blue-black only where a leaf has stored starch, and starch forms only in lit parts. A leaf has three quarters of its area lit and one quarter covered. What fraction of the leaf stays unchanged in the iodine test? [\[Nutrition in Plants\]](#)

- (A) $\frac{2}{3}$
- (B) $\frac{1}{4}$
- (C) 0
- (D) $\frac{3}{4}$
- (E) $\frac{1}{2}$
- (F) $\frac{1}{3}$

42. A bag costs Rs 1000. Its price is first raised by 20 percent, then the new price is cut by 20 percent. What is the final price? [\[Comparing Quantities\]](#)

- (A) Rs 960
- (B) Rs 940
- (C) Rs 1000
- (D) Rs 1200
- (E) Rs 1100
- (F) Rs 1040
- (G) Rs 800

43. Fast-moving air above a flag lowers the pressure there, which makes the flag flutter. If the pressure falls from 1000 units to 960 units, by what percent does it drop? [\[Winds, Storms & Cyclones\]](#)

- (A) 96 percent
- (B) 8 percent
- (C) 4 percent
- (D) 5 percent
- (E) 40 percent
- (F) 2 percent
- (G) 0.4 percent

44. On a clock face, the 12 hour marks are equally spaced around the full circle. What is the angle between the hour and minute hands at exactly 4 o'clock? [\[Lines & Angles\]](#)

- (A) 4 degrees
- (B) 160 degrees
- (C) 150 degrees
- (D) 100 degrees
- (E) 120 degrees
- (F) 30 degrees
- (G) 90 degrees

- 45.** In a food chain grass to grasshopper to frog, only 10 percent of the energy passes on at each step. If the grass holds 5000 units of energy, how much reaches the frog? [\[Nutrition in Plants\]](#)
- (A) 100 units
 - (B) 5000 units
 - (C) 250 units
 - (D) 500 units
 - (E) 50 units
 - (F) 5 units
- 46.** Rs 4000 earns Rs 720 in simple interest over 3 years. What is the rate of interest per year? [\[Comparing Quantities\]](#)
- (A) 12 percent
 - (B) 2 percent
 - (C) 4 percent
 - (D) 9 percent
 - (E) 6 percent
 - (F) 8 percent
 - (G) 18 percent
- 47.** A coastal district issues cyclone warnings: 4 a day for the first 3 days, then 6 a day for the next 2 days. How many warnings are issued in total? [\[Winds, Storms & Cyclones\]](#)
- (A) 30
 - (B) 12
 - (C) 26
 - (D) 10
 - (E) 18
 - (F) 24
- 48.** An angle is three times the size of its complement. What is the measure of the angle? [\[Lines & Angles\]](#)
- (A) 72 degrees
 - (B) 22.5 degrees
 - (C) 75 degrees
 - (D) 67.5 degrees
 - (E) 45 degrees
 - (F) 60 degrees
 - (G) 30 degrees
- 49.** One hectare of forest takes in about 25 tonnes of carbon dioxide a year. A town plants 4 hectares of trees. How much carbon dioxide will they absorb over 3 years? [\[Nutrition in Plants\]](#)
- (A) 100 tonnes
 - (B) 300 tonnes
 - (C) 150 tonnes
 - (D) 500 tonnes
 - (E) 250 tonnes
 - (F) 75 tonnes

50. A 250 g pack of biscuits costs Rs 30, while a 400 g pack costs Rs 44. What does the cheaper pack cost per 100 g? [\[Comparing Quantities\]](#)

- (A) Rs 44
- (B) Rs 14
- (C) Rs 10
- (D) Rs 30
- (E) Rs 11
- (F) Rs 9
- (G) Rs 12

51. Cyclone rain falls at 25 mm per hour. A flat roof of area 8 square metres collects it, where 1 mm of rain on 1 square metre gives 1 litre. How many litres fall on the roof in 3 hours? [\[Winds, Storms & Cyclones\]](#)

- (A) 75 litres
- (B) 150 litres
- (C) 300 litres
- (D) 450 litres
- (E) 600 litres
- (F) 200 litres
- (G) 1800 litres

52. Two angles form a linear pair, and one is 20 degrees more than the other. What is the measure of the larger angle? [\[Lines & Angles\]](#)

- (A) 80 degrees
- (B) 100 degrees
- (C) 120 degrees
- (D) 90 degrees
- (E) 70 degrees
- (F) 160 degrees
- (G) 110 degrees

53. Algae in a pond are producers that double their mass every 2 days. Starting from 5 grams, what is their mass after 6 days? [\[Nutrition in Plants\]](#)

- (A) 80 g
- (B) 15 g
- (C) 35 g
- (D) 30 g
- (E) 20 g
- (F) 40 g
- (G) 25 g

54. A bicycle costing Rs 5000 is sold at a profit of Rs 750. Find the profit percent. [\[Comparing Quantities\]](#)

- (A) 10 percent
- (B) 20 percent
- (C) 7.5 percent
- (D) 12.5 percent
- (E) 15 percent
- (F) 13 percent
- (G) 75 percent

55. Air expands when heated. A 10-litre pocket of air warms up and expands to 13 litres. What is the percentage increase in its volume? [\[Winds, Storms & Cyclones\]](#)

- (A) 33 percent
- (B) 300 percent
- (C) 13 percent
- (D) 23 percent
- (E) 3 percent
- (F) 10 percent
- (G) 30 percent

56. A transversal cuts two parallel lines. Of a pair of co-interior angles, one is twice the other. What is the measure of the smaller co-interior angle? [\[Lines & Angles\]](#)

- (A) 120 degrees
- (B) 30 degrees
- (C) 90 degrees
- (D) 45 degrees
- (E) 60 degrees
- (F) 100 degrees
- (G) 72 degrees

57. Urea fertiliser is 46 percent nitrogen by weight. A farmer spreads 50 kg of urea on a field. How many kilograms of pure nitrogen is that, to the nearest kilogram? [\[Nutrition in Plants\]](#)

- (A) 28 kg
- (B) 23 kg
- (C) 50 kg
- (D) 2.3 kg
- (E) 92 kg
- (F) 4.6 kg
- (G) 46 kg

58. In a survey of 240 people, 35 percent preferred tea. How many people preferred something other than tea? [\[Comparing Quantities\]](#)

- (A) 156
- (B) 120
- (C) 72
- (D) 205
- (E) 84
- (F) 160
- (G) 140

59. Two weather balloons rise straight up, one at 4 m/s and the other at 6 m/s. After 90 seconds, how far apart in height are they? [\[Winds, Storms & Cyclones\]](#)

- (A) 360 m
- (B) 540 m
- (C) 90 m
- (D) 900 m
- (E) 100 m
- (F) 200 m
- (G) 180 m

60. The supplement of an angle is four times the angle itself. What is the measure of the angle? [\[Lines & Angles\]](#)

- (A) 60 degrees
- (B) 20 degrees
- (C) 144 degrees
- (D) 72 degrees
- (E) 30 degrees
- (F) 36 degrees
- (G) 45 degrees

61. A lichen is a partnership in which the alga makes food and the fungus gives shelter and water. If the alga makes 12 units of food, keeps one quarter for itself, and shares the rest with the fungus, how much does the fungus get? [\[Nutrition in Plants\]](#)

- (A) 10 units
- (B) 12 units
- (C) 9 units
- (D) 8 units
- (E) 6 units
- (F) 3 units
- (G) 4 units

62. On a map, 2 cm stands for 5 km on the ground. Two towns are 14 cm apart on the map. What is the real distance between them? [\[Comparing Quantities\]](#)

- (A) 35 km
- (B) 10 km
- (C) 28 km
- (D) 40 km
- (E) 70 km
- (F) 7 km
- (G) 17 km

63. A storm shelter holds 250 people. A coastal town of 4000 people wants shelter space for everyone. How many such shelters are needed? [\[Winds, Storms & Cyclones\]](#)

- (A) 1.6
- (B) 16
- (C) 160
- (D) 14
- (E) 15
- (F) 17
- (G) 20

64. Two adjacent angles together make a right angle. One of them is 18 degrees larger than the other. What is the measure of the bigger angle? [\[Lines & Angles\]](#)

- (A) 18 degrees
- (B) 60 degrees
- (C) 63 degrees
- (D) 54 degrees
- (E) 72 degrees
- (F) 45 degrees
- (G) 36 degrees

65. On a branch of 80 leaves, five eighths have turned brown and lost their chlorophyll. How many leaves are still green? [\[Nutrition in Plants\]](#)

- (A) 16
- (B) 50
- (C) 25
- (D) 30
- (E) 40
- (F) 45
- (G) 8

66. A worker's monthly salary of Rs 20000 is increased by 12 percent. What is the new salary? [\[Comparing Quantities\]](#)

- (A) Rs 2400
- (B) Rs 24000
- (C) Rs 21200
- (D) Rs 17600
- (E) Rs 20012
- (F) Rs 22400
- (G) Rs 22000

67. Strong wind wears away a sand dune at 2.5 cm of height each year. How many years will it take to lower the dune by a full metre? [\[Winds, Storms & Cyclones\]](#)

- (A) 4 years
- (B) 250 years
- (C) 50 years
- (D) 400 years
- (E) 100 years
- (F) 25 years
- (G) 40 years

68. Two lines intersect, and a pair of vertically opposite angles measure $(5x - 10)$ degrees and $(3x + 30)$ degrees. What is the value of x ? [\[Lines & Angles\]](#)

- (A) 25
- (B) 40
- (C) 16
- (D) 20
- (E) 10
- (F) 30
- (G) 8

69. The photosynthesis reaction needs 6 molecules of water for every 1 molecule of glucose made. How many water molecules are used to make 4 glucose molecules? [\[Nutrition in Plants\]](#)

- (A) 6
- (B) 12
- (C) 4
- (D) 30
- (E) 24
- (F) 36

70. Two successive discounts of 10 percent and then 20 percent are given on a Rs 500 item. What is the final amount you pay? [\[Comparing Quantities\]](#)

- (A) Rs 360
- (B) Rs 300
- (C) Rs 375
- (D) Rs 150
- (E) Rs 400
- (F) Rs 350

71. As a storm nears, a barometer falls steadily by 0.8 units every hour. If it now reads 1012 units, what will it read after 5 hours? [\[Winds, Storms & Cyclones\]](#)

- (A) 1012 units
- (B) 1008 units
- (C) 1000 units
- (D) 4 units
- (E) 1016 units
- (F) 1011.2 units
- (G) 1004 units

72. Three angles lie along a straight line and measure 50 degrees, x and 75 degrees. What is the value of x ? [\[Lines & Angles\]](#)

- (A) 45 degrees
- (B) 65 degrees
- (C) 105 degrees
- (D) 50 degrees
- (E) 125 degrees
- (F) 55 degrees

73. Sunlight reaches a leaf for 10 hours, but passing clouds block it 30 percent of that time. For how many hours does the leaf actually photosynthesise? [\[Nutrition in Plants\]](#)

- (A) 30 hours
- (B) 6.5 hours
- (C) 10 hours
- (D) 3 hours
- (E) 7 hours
- (F) 13 hours
- (G) 4 hours

74. If 5 kg of rice costs Rs 240, how much rice can you buy for Rs 600 at the same rate? [\[Comparing Quantities\]](#)

- (A) 10 kg
- (B) 2.5 kg
- (C) 15 kg
- (D) 12.5 kg
- (E) 20 kg
- (F) 12 kg

75. When the air in a hot-air balloon is heated, one fifth of it escapes from the open bottom, making the balloon lighter. If it held 200 kg of air, how much air remains inside? [\[Winds, Storms & Cyclones\]](#)

- (A) 200 kg
- (B) 180 kg
- (C) 160 kg
- (D) 40 kg
- (E) 120 kg
- (F) 100 kg

76. The complement of an angle measures 25 degrees. By how many degrees is the angle itself larger than its complement? [\[Lines & Angles\]](#)

- (A) 50 degrees
- (B) 115 degrees
- (C) 30 degrees
- (D) 65 degrees
- (E) 25 degrees
- (F) 40 degrees
- (G) 90 degrees

77. A sundew plant gets 80 percent of its nitrogen by trapping insects and the rest from the soil. If the plant needs 200 mg of nitrogen, how much comes from the soil? [\[Nutrition in Plants\]](#)

- (A) 160 mg
- (B) 80 mg
- (C) 120 mg
- (D) 100 mg
- (E) 40 mg
- (F) 20 mg
- (G) 180 mg

78. A phone battery loses 20 percent of its charge in the morning, then 30 percent of what remains in the afternoon. What percent of a full charge is left? [\[Comparing Quantities\]](#)

- (A) 60 percent
- (B) 50 percent
- (C) 44 percent
- (D) 80 percent
- (E) 76 percent
- (F) 30 percent
- (G) 56 percent

- 79.** During a cyclone the sea swells up in a storm surge of 3 m above the usual high tide, which is itself 2.5 m above normal sea level. How high does the water reach above normal sea level? [\[Winds, Storms & Cyclones\]](#)
- (A) 6 m
 - (B) 5 m
 - (C) 2.5 m
 - (D) 0.5 m
 - (E) 3 m
 - (F) 4.5 m
 - (G) 5.5 m
- 80.** A clock's minute hand sweeps the full circle in 60 minutes. Through how many degrees does it turn in 20 minutes? [\[Lines & Angles\]](#)
- (A) 120 degrees
 - (B) 72 degrees
 - (C) 60 degrees
 - (D) 20 degrees
 - (E) 90 degrees
 - (F) 100 degrees
- 81.** A potato plant stores food as starch in tubers. Each plant yields 6 tubers averaging 250 g each. What is the total mass of tubers from 20 such plants, in kilograms? [\[Nutrition in Plants\]](#)
- (A) 3 kg
 - (B) 120 kg
 - (C) 30 kg
 - (D) 1.5 kg
 - (E) 15 kg
 - (F) 300 kg
 - (G) 7.5 kg
- 82.** The ratio of red to blue marbles in a jar is 3 : 7. If there are 40 marbles altogether, how many are red? [\[Comparing Quantities\]](#)
- (A) 21
 - (B) 16
 - (C) 18
 - (D) 28
 - (E) 3
 - (F) 12
- 83.** A weather station records the wind speed every 15 minutes, around the clock. How many readings does it take in one full day? [\[Winds, Storms & Cyclones\]](#)
- (A) 60
 - (B) 1440
 - (C) 24
 - (D) 96
 - (E) 48
 - (F) 144
 - (G) 15

- 84.** A transversal cuts two parallel lines so that a pair of corresponding angles measure $(4x)$ degrees and $(2x + 50)$ degrees. What is the measure of one of these angles? [\[Lines & Angles\]](#)
- (A) 120 degrees
 - (B) 100 degrees
 - (C) 50 degrees
 - (D) 25 degrees
 - (E) 75 degrees
 - (F) 130 degrees
- 85.** Carbon dioxide makes up about 0.04 percent of the air. How many litres of carbon dioxide are there in 5000 litres of air? [\[Nutrition in Plants\]](#)
- (A) 200 litres
 - (B) 20 litres
 - (C) 2 litres
 - (D) 0.4 litres
 - (E) 50 litres
 - (F) 4 litres
 - (G) 0.2 litres
- 86.** By selling a watch for Rs 920, a shopkeeper makes a profit of 15 percent. What did the watch cost him? [\[Comparing Quantities\]](#)
- (A) Rs 805
 - (B) Rs 1058
 - (C) Rs 800
 - (D) Rs 850
 - (E) Rs 138
 - (F) Rs 782
- 87.** A weakening cyclone's wind speed drops by 25 percent from 120 km/h. What is the new wind speed? [\[Winds, Storms & Cyclones\]](#)
- (A) 150 km/h
 - (B) 100 km/h
 - (C) 90 km/h
 - (D) 110 km/h
 - (E) 95 km/h
 - (F) 30 km/h
- 88.** An angle measures five sixths of a right angle. What is its size in degrees? [\[Lines & Angles\]](#)
- (A) 150 degrees
 - (B) 108 degrees
 - (C) 75 degrees
 - (D) 54 degrees
 - (E) 15 degrees
 - (F) 60 degrees
 - (G) 90 degrees

- 89.** A plant makes 18 g of glucose in a day and burns two thirds of it for energy, storing the rest. How many grams of glucose does it store? [\[Nutrition in Plants\]](#)
- (A) 9 g
 - (B) 3 g
 - (C) 6 g
 - (D) 12 g
 - (E) 15 g
 - (F) 4 g
 - (G) 18 g
- 90.** A class of 40 students has 50 percent boys. If 10 more boys join, what percent of the new class are boys? [\[Comparing Quantities\]](#)
- (A) 30 percent
 - (B) 75 percent
 - (C) 55 percent
 - (D) 60 percent
 - (E) 50 percent
 - (F) 70 percent
 - (G) 65 percent
- 91.** Rising warm air cools by 1 degree Celsius for every 100 m it climbs. Air leaving the ground at 25 degrees Celsius rises 1500 m. What is its temperature at that height? [\[Winds, Storms & Cyclones\]](#)
- (A) 40 degrees Celsius
 - (B) 20 degrees Celsius
 - (C) 15 degrees Celsius
 - (D) 1500 degrees Celsius
 - (E) 11 degrees Celsius
 - (F) 10 degrees Celsius
- 92.** Two supplementary angles are in the ratio 7 : 5. What is the measure of the smaller angle? [\[Lines & Angles\]](#)
- (A) 150 degrees
 - (B) 60 degrees
 - (C) 90 degrees
 - (D) 75 degrees
 - (E) 100 degrees
 - (F) 105 degrees
 - (G) 84 degrees
- 93.** Nitrogen-fixing bacteria add 8 kg of usable nitrogen to a field each week. Starting from zero, after how many weeks will the field reach 56 kg of added nitrogen? [\[Nutrition in Plants\]](#)
- (A) 8 weeks
 - (B) 14 weeks
 - (C) 9 weeks
 - (D) 7 weeks
 - (E) 5 weeks
 - (F) 6 weeks
 - (G) 48 weeks

94. Find the total amount (principal plus simple interest) on Rs 8000 lent at 5 percent per year for 4 years.

[Comparing Quantities]

- (A) Rs 8400
- (B) Rs 10000
- (C) Rs 1000
- (D) Rs 9600
- (E) Rs 9000
- (F) Rs 1600

95. A coastal region is hit by an average of 3 cyclones every 2 years. At this rate, how many cyclones would it expect over 10 years? [Winds, Storms & Cyclones]

- (A) 5
- (B) 20
- (C) 30
- (D) 9
- (E) 15
- (F) 6

96. Two angles form a linear pair and measure $(x + 25)$ degrees and $(2x - 10)$ degrees. What is the measure of the larger angle? [Lines & Angles]

- (A) 110 degrees
- (B) 100 degrees
- (C) 55 degrees
- (D) 70 degrees
- (E) 90 degrees
- (F) 80 degrees
- (G) 120 degrees

97. Raising the carbon dioxide in a greenhouse boosts photosynthesis, increasing a plant's growth by 35 percent. A tomato plant that would normally grow 40 cm tall now grows how tall? [Nutrition in Plants]

- (A) 94 cm
- (B) 54 cm
- (C) 28 cm
- (D) 60 cm
- (E) 14 cm
- (F) 40 cm
- (G) 75 cm

98. Out of 1500 light bulbs made in a factory, 60 are found to be defective. What percent of the bulbs are defective? [Comparing Quantities]

- (A) 8 percent
- (B) 5 percent
- (C) 6 percent
- (D) 0.4 percent
- (E) 4 percent
- (F) 25 percent
- (G) 40 percent

99. Air pressure pushes on every surface. A window of area 0.5 square metre faces a push of 8000 units on each square metre. What total push does the whole window feel? [\[Winds, Storms & Cyclones\]](#)

- (A) 8000.5 units
- (B) 2000 units
- (C) 8000 units
- (D) 4 units
- (E) 4000 units
- (F) 16000 units
- (G) 800 units

100. An angle measures 110 degrees. What is the measure of its reflex angle? [\[Lines & Angles\]](#)

- (A) 160 degrees
- (B) 250 degrees
- (C) 70 degrees
- (D) 270 degrees
- (E) 180 degrees
- (F) 110 degrees
- (G) 290 degrees